MILITARY SPECIFICATION

BATTERY, WATER ACTIVATED BA-380( )/AMQ-9

1. SCOPE

1.1 This specification covers one type of water activated battery designated as Battery, Water Activated BA-380( )/AMQ-9.

2. APPLICABLE DOCUMENTS

2.1 The following documents of the issue in effect on date of invitation for bids, form a part of this specification:

SPECIFICATIONS

FEDERAL

UU-C-201 — Cardboard, Bristol Manila, and Railroad.

QQ-B-626 — Brass, Leaded and Non-leaded; Rods, Shapes, Forgings and Flat Products with Finished Edges (Bars, Flatwire and Strips)

QQ-C-576 — Copper Plates, Rolled Bars, Sheets and Strips.

QQ-P-330 — Phosphor Bronze Bars, Plates, Rods, Sheets, Strips, Flat Wire, and Structural and Special Shaped Sections.

QQ-S-571 — Solder; Lead Alloy, Tin Lead Alloy, and Tin Alloy; Flux cored Ribbon and Wire, and Solid Form.

PPP-B-636 — Boxes; Fiber.

PPP-C-96 — Cans, Metal, 28 Gage and Lighter.

PPP-C-843 — Cushioning Material, Cellulosic.

PPP-T-76 — Tape, Pressure-Sensitive adhesive paper, water resistant, (for carton sealing).

MILITARY

MIL-W-76 — Wire and Cable, Hook-up Electrical, Insulated.

MIL-P-116 — Preservation, Methods of.

MIL-T-713 — Twine and Tape, Lacing and Tying (for use in Electrical and Electronic Equipment).

MIL-P-3115 — Plastic-Material, Laminated, Thermosetting, Sheets,, Paper base, Phenolic-resin.

MIL-W-3861 — Wire, Electrical (Bare Copper).

MIL-B-10377 — Box, Wood, Cleated, Veneer, Paper Overlaid.

FSC 6135

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MIL-M-13231 — Marking of Electronic Items.

MIL-F-14072 — Finishes for Ground Signal Equipment.

STANDARDS

FEDERAL

Fed-STD-595 — Colors.

MILITARY

MIL-STD-105 — Sampling Procedures and Tables For Inspection by Attributes.

MIL-STD-129 — Marking for Shipment and Storage.

(Copies of specifications, standards and drawings required by contractors in connection with specified procurement functions should be obtained from the procuring activity or as directed by the contracting officer. Both the title and number or symbol should be stipulated when requesting copies.)

2.2 **Publications.** The following documents form a part of this document to the extent specified herein. Unless otherwise indicated, the issue in effect on date of invitation for bids shall apply.

AMERICAN TRUCKING ASSOCIATION

National Motor Freight Classification No. A-3 and No. 14.

(Application for copies should be addressed to the American Trucking Association, 1424 16th Street N. W., Washington, D. C.)

ASSOCIATION OF AMERICAN RAILROADS

(1) Uniform Freight Classification No. 4.

(2) Consolidated Freight Classification Rule No. 21.

(Application for copies should be addressed to Consolidated Classification Committee, 202 Union Station, Chicago, Ill.)

3. REQUIREMENTS

3.1 **Construction.** Battery, Water Activated BA-380( )/AMQ-9, shall be constructed in accordance with the requirements specified herein and in Figure I.

3.2 **Preproduction Samples.** The contractor shall furnish preproduction samples for approval, if and as required by the invitation for bids and contract. (See 4.3 and 6.2(c)(1).)

3.3 **Jacket.** The jacket, (the outer container housing the battery contents) shall be made of plastic material and fabricated in such a manner as to prevent it from coming apart during activation or when subjected to any of the tests specified in Section 4. The jacket shall also be provided with drainage holes.

3.4 **Terminals.** Terminals shall provide a postive means of making and maintaining external electrical connections to the battery. Contact surfaces of the terminals shall not be obstructed by wax or other insulating compounds.

3.4.1 *Socket-type terminals.* Contact portions of socket-type terminals shall be made of spring phosphor bronze or plated spring brass. Sockets shall be of such construction and material as to minimize leakage current between terminals and to prevent the short circuiting of terminals, or even momentary contact of any of the the pins of the mating plugs to any socket terminal other than those for which such pins are intended.

3.5 **Metals.** When used for purposes other than internal cell construction, the metals shall conform as follows:

Metal	*Specification*
Brass	QQ-B-626
Copper	QQ-C-576
Phosphor Bronze	QQ-P-330

3.6 **Parts, Materials, and Processes.**

3.6.1 *Selection.* Parts, materials and processes shall be such as to enable the equipment

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to meet its performance and other requirements, as specified herein.

3.7 **Wire, Hookup.** Conductor size for hookup wire shall be sufficient to avoid deterioration in electrical properties of the wire from overheating.

3.7.1 *Bare.* Bare hookup wire shall not be used unless insulated wire is impracticable because of circuit characteristics or shortness of the wire run. Bare hookup wire shall be type 8, soft or drawn-and-annealed, and coated, and shall conform to MIL-W-3861.

3.7.2 *Insulated.* Insulated wire, when used for electrical connections and terminal leads, shall conform to type LW of MIL-W-76.

3.8 **Cleaning.** Metal parts, after fabrication, shall be cleaned in accordance with good commercial practice, or as specified in an applicable document. Cleaning processes shall have no deleterious effect on the equipment. Corrosive material shall be removed completely before parts are assembled into the equipment. After assembly, components shall be cleaned thoroughly and shall be free from particles of solder, flux, and other foreign material. If necessary, such cleaning shall also be performed before final assembly of components.

3.9 **Finish, Protective.** The equipment shall be given protective finish in accordance with MIL-F-14072. The final paint film on type I surfaces shall be lusterous green, color chip No. 34087 of Federal Standard 595.

3.10 **Marking.** Marking shall conform to MIL-M-18231.

3.10.1 *Terminal Markings.* Polarity, voltage and unit of the battery (A, B, etc.) shall be legibly marked as specified in figure 1.

3.10.2 *Labeling.* Each battery shall be provided with two (2) labels, an identification label and an instruction label, positioned as shown in figure 1. (Notes 3 and 4) The color of lettering shall be black on lusterless green background.

3.10.2.1 *Identification Label.* The identification label shall show the following data:

Battery BA-380( )/AMQ-9
Order Number
Code Number
Manufacturer's name or trade name

3.10.2.1.1 *Code Number.* The code number shall indicate the month and year in which the battery was manufactured by means of a four (4) digit number. The first two digits shall indicate the month and the last two (2) digits the year. Months earlier than the tenth month shall be preceded by "0". When a battery is completed during the last ten (10) working days of a month or the first ten working days of the subsequent month, manufacturer is permitted to use either month as the coded month of manufacture.

3.10.2.2 *Instruction Label.* The instruction label shall contain complete instructions for activating and preparing the battery for service. The text of the instruction label shall be approved by the contracting officer.

3.11 **Soldering.**

3.11.1 *Solder.* Solder used for electrical connections shall be composition Sn60, conforming to QQ-S-571.

3.11.2 *Flux and Cleaning Agents.* No acid or acid salts shall be used in preparation for or during soldering; however, exception is permitted for preliminary tinning of electrical connections and for tinning or soldering of mechanical joints not used to complete electrical circuits, but in no case shall acid or acid salts be used where they can come in contact with insulation material. Where acid or acid salts are used, as permitted above, they shall be completely neutralized and removed immediately after use.

3.11.3 *Process* There shall be no sharp

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points or rough surfaces resulting from insufficient heating. The solder shall feather out to a thin edge, indicating proper flowing and wetting action, and shall not be crystallized, overheated, or under-heated. The minimum necessary amount of flux and solder shall be used for electrical connections. Any means employed to remove an unavoidable excess of flux shall not incur the risk of loose particles of flux, brush bristles, or other foreign material remaining in the equipment; flux being spread over a larger area; or damage to the equipment. Insulation material that has been subjected to heating during the soldering operation shall be undamaged and parts fastened thereto shall not have become loosened.

3.12 **Wiring and Cabling.** Wiring and cabling shall be neat and sturdy.

3.12.1 *Cabling.* Insulated wires shall be formed into cables except where operation of the equipment would be adversely affected thereby, or where it is physically impracticable as in the case where the resulting cables would be excessively large and would interfere with operation or maintenance of the equipment. The cabling of wires shall be effected by twine conforming to MIL-T-713.

3.12.2 *Slack.* Wires and cables shall be as short as practicable except that sufficient slack shall be provided to prevent undue stress on cable forms, wires, and connections; to enable parts to be removed and replaced during servicing without disconnecting other parts; and to facilitate field repair of broken or cut wires.

3.12.3 *Protection.* Wires and cables shall be so placed and protected as to avoid contact, under specified service conditions, with rough or irregular surfaces or sharp edges. Wires shall not be bent sharply where they enter insulation material. Where wires run through holes or slots in metal (partitions, shields, and similar items), they shall be protected by suitable grommets or bushings.

3.12.4 *Connections.* Wire soldered to terminal lugs shall not be bared more than one-eighth of an inch from the lugs, nor extended more than one-eighth inch beyond the lugs.

3.13 **Electrical Requirements.**

3.13.1 *Activation (see 4.8).* The battery shall be capable of being activated as specified in figure 1.

3.13.2 *Initial Voltage-no Load (see 4.9).* After activation in accordance with the instruction label, the minimum voltage for each unit shall be as specified in figure 1.

3.13.3 *Discharge.*

3.12.3.1 *At Low Temperature.* When tested in accordance with 4.10, the voltage of the battery shall not exceed the maximum voltage nor drop below the end voltage specified in figure 1, in less than 145 minutes after initiation of discharge (see 4.10).

3.13.3.2 *At High-Low Temperature.* When tested as specified in 4.11, the voltage of the battery shall not exceed the maximum voltage, nor drop below the end voltage specified in figure 1, in less than 145 minutes after initiation of discharge (see 4.11).

3.13.3.3 *At Normal Temperature.* When tested in accordance with 4.12 the voltage of the battery shall not exceed the maximum, nor drop below the end voltage specified in figure 1, in less than 110 minutes after initiation of discharge.

3.13.4 *Voltage Fluctuation (see 4.13).* At no time during any of the discharge conditions specified in 3.13.3.1 through 3.13.3.3, shall the voltage of the units fluctuate (repetitive voltage changes at intervals of less than 5 seconds duration) beyond the limits specified in figure 1.

3.14 **Flight Simlitude.** Battery, Water Activated BA-380( )/AMQ-9, shall be capable of operation under the simulated flight

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conditions of 4.6. The unit voltages shall not exceed the maximum, nor drop below the end voltages specified in figure 1, in less than 130 minutes after the initiation of the discharge.

3.15 **Workmanship.** The batteries shall be maufactured and assembled in accordance with the applicable portions of the following paragraphs herein:

3.6	Plastic material and plastic parts
3.8	Cleaning
3.11.2	Flux and cleaning agents for soldering
3.11.3	Process for soldering
3.12	Wiring and cabling
3.12.2	Slack in wires and cables
3.12.3	Protection for wires and cables
3.12.4	Connections

4. QUALITY ASSURANCE PROVISIONS

4.1 **Responsibility for Inspection.** Unless otherwise specified in the contract or purchase order, the supplier is responsible for the performance of all inspection requirements as specified herein. Except as otherwise specified, the supplier may utilize his own facilities or any commercial laboratory acceptable to the Government. The Government reserves the right to perform any of the inspections set forth in the specification where such inspections are deemed necessary to assure that supplies and services conform to prescribed requirements.

4.2 **Classification of Inspection.** Inspection shall be classified as follows:

(a) Preproduction inspection (does not include preparation for delivery) (see 4.3).

(b) Inspection covered by subsidiary documents (see 4.4).

(c) Quality conformance inspection.

(1) Quality conformance inspection of equipment before preparation for delivery (see 4.5).

(2) Quality conformance inspection of preparation for delivery (see 4.19).

4.3 **Preproduction Inspection.** Preproduction inspection shall consist of the examination and tests specified in table I. This inspection will be performed by the Government, except for the flight similitude test, which will be performed by the supplier on five (5) sample batteries. Other non-destructive tests on preproduction samples may be performed to determine compliance with specified requirements.

TABLE I. *Preproduction Inspection*

Inspection	Req. para.	Insp. para.
Visual and Mechanical	3.15	4.18
Activation	3.13.1	4.8
Initial Voltage	3.13.2	4.9
Low Temperature Discharge	3.13.3.1	4.10
High-Low Temperature Discharge	3.13.3.2	4.11
Normal Temperature Discharge	3.13.3.3	4.12
Flight Similitude Discharge	3.14	4.6
Voltage Fluctuation (performed simultaneously with the above discharge tests)	3.13.4	4.13

4.3.1 *Definition of Failure.* If any of the preproduction batteries fail to pass these inspections, the preproduction sample lot will be considered unacceptable.

4.3.2 *Procedure in Case of Failure.* The contractor, upon formal notification that the preproduction samples have failed, shall immediately investigate the cause or causes of

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failure and take corrective action to assure that a subsequent lot or lots, upon test, do not contain the same defect or defects. The contractor shall not begin production until a preproduction lot has satisfactorily passed this inspection.

4.4 Inspection Covered by Subsidiary Documents. The following shall be inspected under the applicable subsidiary documents as part of the acceptance inspection before preparation for delivery.

Item	*Where required*
Finish	3.9
Marking	3.10

4.5 Quality Conformance Inspection of Batteries before Preparation for Delivery. Quality conformance inspection of batteries before preparation for delivery shall consist of Groups A, B, C, and D. The contractor shall perform the inspection specified in 4.4, 4.5.2, 4.5.3, and 4.5.5. The inspection specified in 4.5.4 shall be performed by the Government. This does not relieve the contractor of his responsibility for performing any additional inspection which is necessary to control the quality of the product and to assure compliance with all specification requirements. The Government will review and evaluate the contractor's inspection procedures and examine the contractor's inspection records. In addition, the Government at its discretion, may perform all or any of the specified inspections, to verify the contractor's compliance with specified requirements (see 6.3). Test equipment for Government verification inspection shall be made available by the contractor. A battery lot that has passed Groups A and B inspection shall be acceptable for shipment by the contractor. The shipment of a lot of batteries shall not be held up pending results of Groups C and D inspection performed on samples representing the lot.

4.5.1 *Lot.* All batteries of the same type designation offered for delivery during any one month on any one contract or order, shall be considered a lot (quantity on release) for purpose of sampling and inspection.

4.5.1.1 *Sub-lot.* At the supplier's option, sampling and inspection may be performed more frequently on a smaller-than-lot (sub-lot) basis. The group A sample size for a sub-lot shall be determined in the same manner as that for a lot, using the size of the sub-lot submitted and the inspection level specified (see 4.5.2). The Group B sample size for a sub-lot shall be determined in the same manner as that for a lot, using the size of the sub-lot sumbitted and the inspection level specified (see 4.5.3). The Group C sample size for a sub-lot shall have the same ratio to the lot sample size that is specified in table V (see 4.5.4.1) as the sub-lot has to the lot (Quantity or release).

4.5.2 *Group A Inspection.* This inspection including sampling, shall conform to Table II, and the ordinary inspection procedures of Standard MIL–STD–105. Group A inspection shall be performed in any order which is satisfactory to the Government.

TABLE II. *Group A Inspection*

Inspection	Req. para.	Insp. para.	AQL	
			Major	Minor
Visual and Mechanical	3.15	4.18	1%	4%

4.5.3 *Group B Inspection.* This inspection, including sampling, shall conform to table III, and to the special procedures for small-sample inspection of MIL–STD–105. The sample size shall be in accordance with special inspection level "S–4" of MIL–STD–

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105. Group B inspection shall normally be performed on inspection lots that have passed Group A inspection and on samples selected from units that have been subjected to and met the Group A inspection.

TABLE III. *Group B Inspection*

Inspection	Req. para.	Insp. para.	AQL
Activation	3.13.1	4.8	
Initial Voltage	3.13.2	4.9	
Flight Similitude Discharge	3.14	4.6	6.5%
Voltage Fluctuation (performed simultaneously with above discharge test)	3.13.4	4.13	

4.5.4 *Group C Inspection.* Group C inspection shall consist of the delayed capacity and voltage tests in Table IV, and shall be performed by the Government.

TABLE IV. *Group C Inspection*

Test	Storage time	Req. para.	Insp. para.
Low Temperature Discharge	12 Months	3.13.3.1	4.10
High-Low Temperature Discharge	12 Months	3.13.3.2	4.11
Normal Temperature Discharge	12 Months	3.13.3.3	4.12

4.5.4.1 *Sampling for Inspection.* Samples for Group C inspection shall be selected from an accepted lot and shall be as specified in table V.

TABLE V. *Sample Batteries for Group C Inspection*

Quantity on release	Number of sample batteries
Up to 90	5
91 to 150	8
151 to 500	13
501 to 1200	20
1201 to 10,000	32
10,001 to 35,000	50
35,001 to 500,000	80
500,001 and over	125

4.5.4.2 *Marking and Storage, Sample Batteries.* All batteries which have been selected for Group C inspection shall be stored in their shipping containers at the temperature of 70°F. ± 5°F., and relative humidity of 50% ± 15%, until shipment is made. All packages, cartons, and cases containing sample batteries shall be marked: DO NOT OPEN — TEST SAMPLE.

4.5.4.3 *Shipment.* Sample batteries selected for Group C inspection shall be shipped to the Government inspection facility as specified herein, at the contractor's expense.

U. S. Army Electronics Command
Procurement & Production Directorate
ATT: Test Equipment & Battery Division
Squier Area, Building T–292
Fort Monmouth, New Jersey 07703

4.5.4.4 *Non-Compliance.* If the Group C sample batteries representing a particular lot fail to meet performance criteria specified in 3.13, the contract will be subject to adjustment. The value of adjustment shall be determined in accordance with 6.4.

4.5.5 *Group D Inspection.* Group D inspection shall consist of the flight similitude discharge test and is to be performed by the supplier in plant under the conditions as specified herein.

4.5.5.1 *Sampling.* Sample batteries for Group D inspection shall be selected at ran-

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dom from the quantity on release by the Governmnent representative in accordance with 4.5.4.1.

4.5.5.2 *Test Routine.* The sample batteries as selected shall be stored for a period of 12 months at 70°F., ± 5°F., and 50% ± 15% RH, at the supplier's plant, and after storage the batteries shall be tested in accordance with 4.5 and witnessed by the Government representative in residence.

4.5.5.3 *Test Results.* Three (3) copies of the test results shall be furnished, as follows:

U. S. Army Electronics Command
Procurement & Production Directorate
Test Equipment & Battery Division
ATTN: AMSEL-PP-E-TEB-8
Fort Monmouth, New Jersey 07703

4.6 **Flight Similitude Test.** After the battery has been activated in accordance with manufacturer's instructions and has reached the minimum initial voltage, the battery shall then be inserted in radiosonde AN/AMQ-9 or a substitute box having the same insulating properties as the radiosonde. The radiosonde or the substitute box, with the activated battery mounted therein, shall be placed in a test chamber at ambient room temperature and ambient atmosphere pressure. Simutaneously with the beginning of the discharge, the pressure and the temperature shall be reduced in accordance with table VI. Temperature, pressure and voltage of each unit shall be read at 5 minute intervals (see 3.14).

TABLE VI. *Time, Temperature, Pressure for Simulated Flight Testing*

TIME	ALTITUDE	PRESSURE	TEMPERATURE
From instant of simulated-flight initiation (Tolerance: ± 1 minute) Minutes	(simulated) Above sea level Feet	In the test chamber (Tolerance: ± 10% of the listed pressure) Inches of Hg.	In the test chamber (Tolerance: ± 5°F.) °F.
0.0	0	29.92 (Approx)	+59.0
5.0	5,000	24.90	+41.2
10.0	10,000	20.58	+23.3
15.0	15,000	16.89	+ 5.5
20.0	20,000	13.75	—12.3
25.0	25,000	11.10	—30.2
30.0	30,000	8.89	—48.0
35.0	35,000	7.04	—65.8
40.0	40,000	5.54	—69.7
45.0	45,000	4.36	—69.7
50.0	50,000	3.42	—69.7
55.0	55,000	2.69	—69.7
60.0	60,000	2.12	—69.7
65.0	65,000	1.67	—69.7
70.0	70,000	1.31	—69.7
75.0	75,000	1.03	—69.7
80.0	80,000	0.81	—69.7
85.0	85,000	0.64	—64.2
90.0	90,000	0.50	—56.6
95.0	95,000	0.40	—48.8
100.0	100,000	0.32	—40.1
105.0	105,000	0.26	—31.9
110.0	110,000	0.21	—23.6
115.0	115,000	0.17	—15.4
120.0	120,000	0.14	— 7.2
125.0	125,000	0.11	+ 1.0
130.0	130,000	0.09	+ 9.3

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4.7 **Resistance Loads.** For any of the discharge tests, the battery shall be discharged through the following resistance loads:

Unit	*Resistance*
A_1	3.0 ± .06 ohms
A	87.5 ± .14 ohms
B	2347.0 ± 15.0 ohms

4.8 **Activation.** The batteries shall be activated as specified on the instruction label. All activation and preparation shall be accomplished at a temperature of 70°F. ± 5°F. The total time required for activation and preparation shall not exceed that specified in figure I for any battery. During the first 10 minutes of activation, the battery shall have no external load applied. During the remaining 15 minutes, the battery shall be discharged through the resistance loads specified in 4.7. Immediately upon completion of the 15 minute discharge, the battery's voltage shall be tested as specified in 4.10.

4.9 **Initial Voltage.** The batteries shall be tested to determine compliance with 3.13.2.

4.10 **Discharge at Low Temperature.** After activation, the batteries shall be discharged through the resistance loads specified in 4.7, at a temperature of —58°F. ± 2°F. to determine compliance with 3.13.3.1.

4.11 **Discharge at High-low Temperature.** After activation, the batteries shall be discharged through the resistance loads specified in 4.7, at a temperature of +140°F. ± 2°F., for a time period of 10 minutes and then at a temperature of —58°F. ± 2°F., for the remaining time to determine compliance with 3.13.3.2.

4.12 **Discharge at Normal Temperature.** After activation, the batteries shall be discharged through the resistance loads specified in 4.7, at a temperature of 80° ± 10°F., to determine compliance with 3.13.3.3.

4.13. **Voltage Fluctuation.** While performing any of the discharge tests specified in 4.6 and 4.9 through 4.12, the voltage shall be tested to determine compliance with 3.13.4.

4.14 **Standard Test and Storage Conditions.**

4.14.1 *Test Conditions.* Unless otherwise specified, all tests shall be conducted at temperature of 75°F. ± 10°F. (23.9°C. ± 5.6°C.).

4.14.2 *Storage Conditions.* All batteries intended for capacity and voltage tests shall be stored in their shipping containers at the temperature and relative humidity specified within 2 weeks after date of receipt at the designated laboratory.

4.15 **Test Equipment and Inspection Facilities.** Test equipment and inspection facilities shall be of sufficient accuracy and quality to permit performance of the required inspection. The manufacturer shall establish adequate calibration of test equipment to the satisfaction of the Government.

4.15.1 *Instrument Accuracy.*

4.15.1.1 *Voltmeters.* All batteries subjected to a test load shall be read with a voltmeter having a resistance of at least 1,000 ohms per volt. All voltmeters used shall be accurate within 1% of the full scale reading. The voltmeter range shall be such that the readings are taken on the upper half of the scale.

4.15.1.2 *Resistor Tolerances.* In all tests involving discharge through a resistance, such resistance shall be of the following accuracy:

(a) Resistances from zero (0) to 25,000 ohms shall be accurate within 0.5%.

(b) Resistances from above 25,000 ohms to and including 1 megohm shall be accurate within 1%.

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(c) Resistances above 1 megohm shall be accurate within 5%.

(d) In determining the resistance used as a test load, the resistances of all continuously operating voltmeters used shall be considered as part of the specified resistance load (see 4.7).

4.15.1.3 *Timing*. Timing equipment shall be accurate within 0.5%.

4.16 **Battery Container.** All batteries that are subjected to discharge tests shall be tested in the Radiosonde AN/AMQ-9 battery compartment or in a substitute box which offers equal insuating properties and resistive loads.

4.17 **Test Chamber Conditions.** The test chamber required for the discharge tests shall be capable of:

(a) Simulating temperatures specified

(b) Provide air flow moving, as follows: During low temperature discharge tests and the low temperature phase of the high-low temperature discharge tests, the air motion within a proximity of 2 inches to 6 inches of the battery shall be maintained in the range of 40 to 100 feet per minute.

4.18 **Visual and Mechanical Inspection.** Batteries shall be inspected to verify that the materials, external and internal construction (including sockets or terminals, physical dimensions, weight, marking, and workmanship) are in accordance with the applicable requirements.

4.19 **Quality Conformance Inspection of Preparation for Delivery.** Preparation for delivery shall be inspected in accordance with MIL-P-116 to determine conformance to the requirements of Section 5.

5. PREPARATION FOR DELIVERY

5.1 **Preservation and Packaging.**

5.1.1 *Level A*. Battery BA-380( )/AMQ-9 shall be preserved and packaged in accordance with the procedures specified for the designated method prescribed in MIL-P-116, as described in the following paragraphs.

5.1.1.1 *Unit Packaging*. Battery BA-380( )/AMQ-9, shall be individually packaged Method 1A7, as follows: Cushion each Battery by wrapping with ½ inch thickness, Type II, Class B, cellulose wadding conforming to PPP-C-848. The cushioned Battery shall be snugly inclosed within a Type III metal can with key attached conforming to PPP-C-96. Plan B exterior coating with olive drab finish shall be required.

5.2 **Packing.**

5.2.1 *Level A*. A quantity of 8 batteries, BA-380( )/AMQ-9, packaged as specified in 5.1.1.1, shall be packed for shipment within a fiberboard box conforming to Type I, Class 2, Grade 8, of PPP-B-636. Box closure shall be as specified in the appendix of the box specification. In addition, seams and manufacturer's joints shall be sealed with minimum 2 inch wide tape conforming to PPP-T-76.

5.2.1.1 *Metal Strapping*. Shipping containers shall be strapped in conformance with the appendix of the box specification.

5.3 **Marking.** Interior packages and exterior shipping containers shall be marked in accordance with the applicable provisions of MIL-STD-129.

5.4 **Package Testing.** Interior packages will be tested to assure compliance with the requirements specified herein.

5.5 **Workmanship.** The quality of work-

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manship shall be such as to permit acceptance of the completed preservation, packaging, packing and marking requirements in accordance with the testing specified in 4.19.

6. NOTES

6.1 Intended Use. Battery, Water Activated BA-380( )/AMQ-9, is used to provide power for Radiosonde AN/AMQ-9.

6.2 Ordering Data. Procurement documents shall specify the following:

(a) Title, number and date of this specification and any amendment thereto;

(b) Level of packaging and level of packing required for shipment; (Level A, Level B, or Level C)

(c) Preproduction inspection;

(1) Thirty-one (31) samples are generally required

(2) Preproduction pack(s) as follows:

(a) Makeup of pack(s)

(b) Number of each kind of pack to be submitted

(c) Inspection to be performed thereon

(d) Marking and shipping of samples;

(e) Place of final inspection.

6.3 Verification Inspection. Verification by the Government will be limited to the amount deemed necessary to determine compliance with the contract and will be limited to severity to the definitive quality assurance provisions established in this specification and the contract. The amount of verification inspection by the Government will be adjusted to make maximum utilization of the contractor's quality control system and the quality history of the product.

6.4 Adjustment. If as a result of Group C inspection, it is determined that the normal temperature, the low temperature or the high-low temperature discharge tests on sample batteries representing a particular lot fail to meet the performance requirements specified herein, the particular lot in question shall be subject to adjustment. The adjustment shall be based on the discharge test (if more than one type discharge tests fail) that has the greater number of defectives. If more than one lot is found to be subject to adjustment, the total adjustment shall be the sum total of all lot adjustments each calculated separately.

6.4.1 *Contract Adjustment.* The following formula shall be used to determine contract adjustment.

(a) CA (in batteries) = $(\frac{d}{n} - .065)\ N$

(b) CA (in dollars) = $(\frac{d}{n} - .065)\ NP$

CA = Contract Adjustment

d = Number of Defective Sample Batteries

n = Number of Sample Batteries involved in the Test

N = Number of Batteries represented by the Sample n

P = Contractual cost of each Battery

If contract adjustment is found necessary, this adjustment shall not exceed the calculated value of 0.7 N, or 0.7 NP, as applicable.

6.4.2 *Procedure of Adjustment.* The contractor shall be notified in writing of the failure within six (6) months after completion of inspection. Adjustment shall be made in cash or in batteries, all such additional batteries should conform to all requirements of this specification and should be subject to adjustment as specified herein. However, these adjustments should be made in cash.

6.4.3 *Value of Adjustment.* On those ship-

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ment releases in which adjustment is necessary, the adjustment shall be determined by the formula a or b, paragraph 6.4.1.

6.4.3.1 *In Cash.* The total adjustment made in cash shall be equal to the sum of all shipment release adjustments.

6.4.3.2 *In Batteries.* The total adjustment made in batteries shall be equal to the total adjustment in cash divided by the contract price of each battery.

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FIGURE 1. *Electrical and Physical Requirements*

UNIT	NOMINAL	VOLTAGE OF UNIT				FLUCTUATION (1)
		MAXIMUM DURING DISCHARGE AT		MINIMUM		
		High and Low Temperature	Normal Temperature	After Activation	Test end	
A_1	1.5	1.5	1.7	1.1	1.099	± 0.01
A	6.0	7.4	7.9	5.5	5.499	± 0.05
B	115.0	125.0	130.0	95.0	94.99	± 0.50

Time for Activation:

On Open Circuit	10 Minutes
On Closed Circuit	15 Minutes
Total activation, Maximum	25 Minutes

Resistance Load

A_1 Unit	8.0 ± 1.06 ohms
A Unit	37.5 ± .14 ohms
B Unit	2347.0 ± 15.0 ohms

Terminals	Socket
Weight (activated) maximum	600 grams

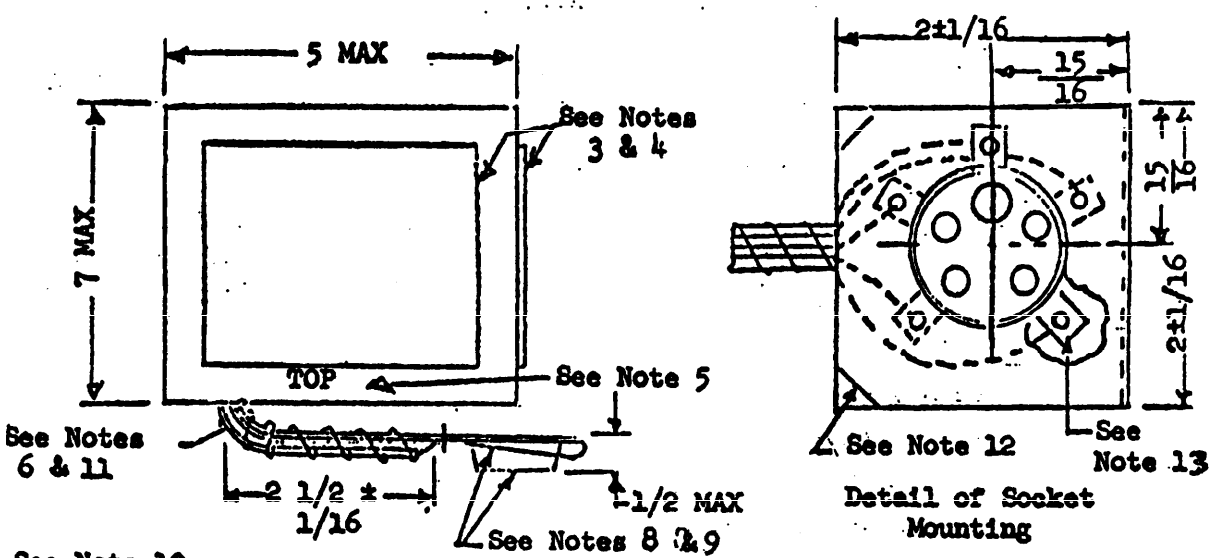
Capacity and Voltage Test (minimum time to reach test end voltage)

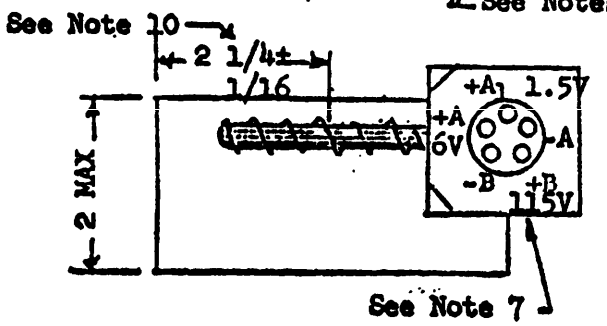
Low Temperature Discharge	145 Minutes
High-Low Temperature Discharge	145 Minutes
Normal Temperature Discharge	110 Minutes
Flight Similitude Discharge	180 Minutes

(1) During any of the discharge tests.

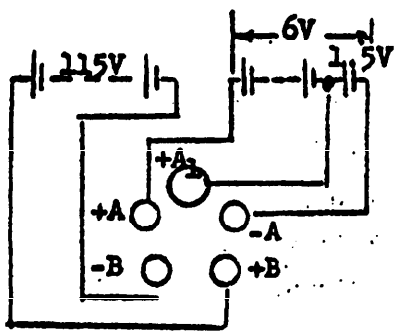
FIGURE I. *Sheet 1 of 3*

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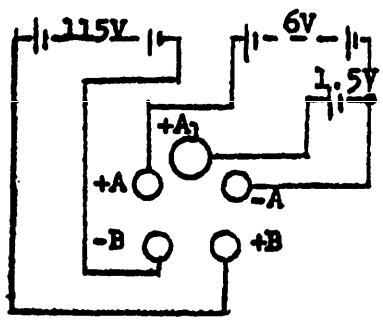




Weight: Battery activated in accordance with the manufacturer's instruction shall not exceed 600 grams.



Schematic Diagram



← Alternative Schematic Diagram

FIGURE I. *Sheet 2 of 3*

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NOTES

1. All dimensions in inches.

2. Unless otherwise specified, tolerance is ± 1/32.

3. Identification label, centered on side of battery.

4. Instruction label, centered on top of battery.

5. Permanent black 3/16 character, centered.

6. Hook-up wire conforming to MIL-W-76 shall be as follows:

TERMINAL	TYPE
—A	LW-C-24(7)-J-0
$+A_1$ 1.5 v	LW-C-24(7)-J-4
+A 6 v	LW-C-24(7)-J-2
—B	LW-C-24(7)-J-6
+B 115 v	LW-C-24(7)-J-8

7. Cardboard, faced on both sides, conforming to Class II, white, 8 ply, .030 thick, of Specification UU-C-201.

8. Backing plate of socket conforming to type PBE-P of Specification MIL-P-8115

9. Five-hole socket No. 9705-23, H. H. Eby, Inc., or equal.

10. Wire outlet shall be located within this area.

11. Wires may be twisted or laced with twine conforming to Type N, Class 3, of Specification MIL-W-713.

12. Two staples, Pilot No. 400, Ace Fastener Corporation, or equal; or equivalent means of fastening.

13. Leads to be securely soldered to terminal lugs.

14. Referenced specifications shall be of the issue in effect on date of invitation for bids.

FIGURE 1. *Sheet 3 of 3*

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SPECIFICATION ANALYSIS SHEET

Form Approved Budget Bureau No. 119-R004

INSTRUCTIONS

This sheet is to be filled out by personnel either Government or contractor, involved in the use of the specification in procurement of products for ultimate use by the Department of Defense. This sheet is provided for obtaining information on the use of this specification which will insure that suitable products can be procured with a minimum amount of delay and at the least cost. Comments and the return of this form will be appreciated. Fold on lines on reverse side, staple in corner, and send to preparing activity.

SPECIFICATION

ORGANIZATION | CITY AND STATE

CONTRACT NO. | QUANTITY OF ITEMS PROCURED | DOLLAR AMOUNT $

MATERIAL PROCURED UNDER A

☐ DIRECT GOVERNMENT CONTRACT ☐ SUBCONTRACT

1. HAS ANY PART OF THE SPECIFICATION CREATED PROBLEMS OR REQUIRED INTERPRETATION IN PROCUREMENT USE?
   A. GIVE PARAGRAPH NUMBER AND WORDING

   B. RECOMMENDATIONS FOR CORRECTING THE DEFICIENCIES

2. COMMENTS ON ANY SPECIFICATION REQUIREMENT CONSIDERED TOO RIGID

3. IS THE SPECIFICATION RESTRICTIVE?

   ☐ YES ☐ NO IF "YES", IN WHAT WAY?

4. REMARKS *(Attach any pertinent data which may be of use in improving this specification. If there are additional papers, attach to form and place both in an envelope addressed to preparing activity)*

SUBMITTED BY *(Printed or typed name and activity)* | DATE

DD FORM 1426 1 APR 63 REPLACES NAVSHIPS FORM 4853, WHICH IS OBSOLETE

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